Formal Epistemic Structures and Mechanics — V5

Working framework draft

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Base manuscripts: `FINAL_V2_1.md`, `FINAL_V3.md`, `FINAL_V4.md` (historical scientific layers retained)

V4 presented a certificate-lifting semantics whose support was an exhaustive enumeration over a bounded model. V5 changes what the paper claims to have proved, in both directions. Eleven theorems are now discharged over uninterpreted sorts, so the separation and change-propagation results hold at any size rather than at the enumerated one; and three of the quantities V4 reported as findings are shown here to have been unfalsifiable, so they are withdrawn as evidence and kept only as a record of what the enumeration could and could not have detected.

1 Replacement abstract for V5

Systems that carry out scientific work inside mature governance machinery accumulate certificates: runtime proof-of-execution, certified proposal–execution traces, portable action and approval receipts, workflow provenance signatures, dependency and effect systems, attested execution boundaries. This paper treats those mechanisms as reusable donor certificates rather than competitors, and asks what it takes to lift one of them into continued *scientific* standing after the scientific state changes.

The answer is a conservative lifting semantics with two halves, and both are now proved rather than enumerated. The **local half** governs what a change touches: separated mechanics commute, separation is necessary for commutation rather than merely sufficient, writes are local to a declared frame, the frame determines the result, and fixpoints are conservative. The **propagation half** governs what a change invalidates: reopening is sound, complete and minimal, conservative and monotone, and a node reopens itself exactly when it lies on a cycle. Both halves are discharged over uninterpreted sorts — the state transformer of the local half is an uninterpreted function under a frame condition, and the propagation half quantifies over dependency graphs of any size, cyclic or not — so neither is a statement about a particular width.

The bounded model of V4 is then recovered as an instance rather than repeated as separate support. A certificate over n coordinates is interpreted as the star graph on $n+1$ nodes: one node per coordinate, one for the certificate, an edge from each coordinate to it. Damage is a changed set on the coordinate nodes and repair removes coordinates from it. Under that interpretation, six further theorems are discharged at arbitrary width — any damage withdraws the certificate, a full repair restores it, a partial repair does not, undamaged coordinates are never collateral damage, the dependency runs one way, and the certificate supports nothing — and the two revalidation counts V4 published are recomputed by running the implementation the propagation theorems were checked against. The finite result is therefore output of a verified instance of the theorems, not a second enumeration that agrees with them.

Three of V4's reported quantities do not survive this scrutiny and are withdrawn below.

2 What the enumeration could not have found

The bounded model of V4 reported seven quantities. Three of them could not have come out any other way, and one of the three had never been evaluated at all. Stating this is not a concession extracted after the fact; it is the result of pointing the mutation discipline that validates the new theorems at the old checks.

The change-propagation check asserted set-algebra tautologies. It computed a downstream set and then asserted two things about it: that the retained set does not intersect the downstream set, and that the retained set equals the certified set minus the downstream set. The first intersects a set with something already removed from it. The second compares a variable to the expression it was just assigned. Both hold for *any* value of the downstream set. The propagation function was called and its output was never compared against anything. Confirmed by mutation: replacing it with "always the empty set" left the check passing, and so did "always every node", both reporting the same case count as the real implementation. The reported 130,320 cases over 543 acyclic graphs were therefore 130,320 cases none of which could fail.

The check itself has since been repaired, on 2026-08-22, and the sentence above is a description of what it was. Its assertions now compare the retained set against a specification built from an independently computed transitive closure, so both mutants die where both previously survived, and eight declared wrong propagation operators are rejected where all eight were accepted. The case count is unchanged at 130,320 over 543 graphs; what changed is that those cases can now fail.

That is now repaired rather than merely reported. The propagation specification is written independently of the implementation, and the shipped implementation is compared against it exhaustively: **61,440 of 61,440** directed graphs on four nodes and changed subsets agree, **51,712** of them with a non-empty reopening. The implementation is correct. Before this, that had not been established.

Donor conservativity could not be violated. The counter assigns a projected verdict from the donor-native verdict on one line and compares the two on the next. The condition is `x != x`. No theory of lifting, however wrong, moves that count off zero. It is withdrawn as evidence; donor conservativity is instead carried by the fixpoint-conservativity theorem, which is proved.

Ideal-product equivalence compares an expression to itself. The ideal product is defined as the same predicate the lifting rule is defined as, so the mismatch count is zero by construction. The equivalence claim is real and worth making — an information-equivalent product carrying the same coordinates and the same lift predicate does agree extensionally — but it is a definitional identity and the enumeration adds nothing to it. It is withdrawn as an empirical quantity.

The "independent" reproduction is a paraphrase. The second implementation replaces a universal quantifier with an early-return loop. It diverges from the reference on 0 of 320 points because it cannot diverge on any. A second implementation that cannot disagree confirms a transcription, not a theorem. The claim of independent reproduction is withdrawn; genuine independent review remains outstanding and is named in the limits below.

And one wrong theory walks through every check that remains. Each assertion in the bounded checker evaluates the lift rule with the donor certificate valid. The 32 points of the space where it is invalid are enumerated into the published digest and never asserted about. A rule that lifts on the scientific coordinates alone, ignoring the donor entirely, passes the whole battery. Conservativity is carried by the 1,536-state derivation reported below, where dropping it disagrees on 72 states — not by the bounded checker.

3 The general theorems

Local mechanics (P6.COMMUTE.RW_NONINTERFERENCE.V1). Let deterministic, admissible mechanics act on a state over uninterpreted coordinate and value sorts. Each mechanic must be read-footprint faithful and write-footprint faithful. Their write coordinates are distinct, neither mechanic reads the coordinate written by the other, and neither changes an obligation, authority fact, provenance object, dependency, resource, or declared external input consumed by the other. Whenever both sequential compositions are defined, the two orders have the same current scientific projection. Their ordered audit histories need not be literally equal; they are equivalent under swaps of adjacent independent commit events.

The two cross-read exclusions are load-bearing. If either is removed while the writes remain disjoint, a frame-faithful transformer and state can make the two orders disagree. This is a necessity claim about what the stated assumptions must exclude to entail commutation for every admissible transformer, not a claim that every particular pair of interfering mechanics fails to commute. The bounded V4 modular check is only an instance of this stronger contract, and the shipped mechanic obeys its declared frame on all **5,280** exhaustive perturbations.

Change propagation. Over an uninterpreted node sort, with reachability axiomatised as the reflexive–transitive closure of the dependency edge: reopening is sound (every reopened node is reachable from a change), complete (every reachable non-changed node is reopened), minimal (no proper subset of the reopened set is closed under the edge relation from the changed set), conservative, monotone in the changed set, and a node lies in its own reopened set exactly when it lies on a cycle. Graphs of any size, cyclic or acyclic.

One clause of the closure axiomatisation is load-bearing and easy to omit. Transitive closure is not first-order definable, so the four clauses *are* the definition, and the fourth attaches a well-founded rank to each reachability fact. Without it a model may contain two reachability facts that justify each other in a cycle, and the completeness direction becomes unprovable. That was not anticipated: an earlier axiomatisation without the rank admitted a model in which a sink domain reached its own source, and a differential against the committed implementation is what exposed it.

3.1 The exact commutation statement, kernel-mechanized

The full Theorem 7 statement – multi-component environments, mechanics that are read-footprint faithful and write-footprint faithful over their declared footprints, and fully scientifically separated – is machine-checked as a kernel proof under contract id P6.COMMUTE.EXACT_THEOREM7.V1. Every derivation step is an application of a rule in a fixed small kernel, the serialized proof is replayed from nothing in a fresh kernel, and the replay reproduces the exact conclusion with every residual hypothesis inside the declared theory; a z3 cross-check of the same sentence runs alongside. The conclusion pairs equality of the two scientific projections with swap-equivalence of the two ordered histories under independent events. The kernel is programme-authored Python rather than an external proof assistant, and that boundary is stated in the mechanized record.

4 The bounded model as an instance

Two derivations connect the general theorems to what V4 published.

The lift rule. Three primitives are stated without reference to the rule: a donor certificate establishes only what its own embedding declares; each scientific coordinate is a separate obligation rather than a contribution to a score; and lifting is conservative, adding no authority the donor did not have. The lift rule computed from them reproduces the shipped rule on all **1,536** states of the embedding cube, with both verdicts present (24 admissible, 1,512 not). It is a derivation and not a restatement: making a scientific coordinate compensatory disagrees on 96 states, waiving an embedding's donor requirement on 9, and dropping conservativity on 72.

The revalidation counts. A certificate over n coordinates is the star graph on $n+1$ nodes; damage is a changed set on the coordinate nodes; repair removes coordinates from it. Six theorems are discharged under that reading at arbitrary n, listed in the abstract above. The two published counts — **155** full restorations and **1,055** proper-subset failures — are then recomputed by running the shipped propagation implementation over the interpreted graph, so they are output of an implementation independently verified against the theorems rather than a separate enumeration agreeing with them. Independently, the shipped revalidation assertion block accepts the derived rule and rejects three weakenings of the non-compensatory primitive.

Two properties of that pair are stated because they are not obvious from the numbers.

The 155 discriminates between almost nothing. A full repair leaves nothing changed, which satisfies any monotone admissibility rule and reopens nothing in any dependency graph. Every wrong graph tried returns 155, and so does every weakened primitive tried. It is the number of (donor, non-empty damage) pairs. The 1,055 is the count that moves: to 655, 255 and 0 under the three primitive weakenings.

And neither count identifies the interpretation. This is the sharper limitation, and it was found by looking for it. Seven dependency graphs were tried against the star. Three collapse the failure count to zero — edges reversed, no edges, coordinates chained without reaching the certificate. One moves it to 975 — a single coordinate stops supporting the certificate. But three reproduce **1,055 exactly**: a chain running through the coordinates *into* the certificate, the star with extra coordinate-to-coordinate edges, and the complete graph. The pair of counts tests whether every coordinate reaches the certificate, and nothing beyond that; every graph in that reachability class returns the same two numbers.

What separates the star from those three is the proof, not the arithmetic. Each of them reopens coordinates as collateral damage — 245, 100 and 375 instances respectively — and collateral reopening is precisely what *undamaged coordinates are never collateral damage* forbids, and precisely what is lost when the frame condition forbidding coordinate-to-coordinate support is dropped. The shipped implementation is confirmed to reopen nothing but the certificate under the star. So the theorems pin the graph and the counts confirm its reachability class; a paper that reported the counts alone as evidence for the structure would be reporting something the counts cannot see.

The donor axis is a multiplier, not a dimension. The donor family enters neither the dependency graph nor the changed set. The five families replicate the same 31 restorations and 211 failures rather than extending them: 155 and 1,055 are 31 and 211 counted five times. The same loop repeats the rest of the list: 320 states is the 64-point (donor verdict, coordinate vector) space enumerated once per family, and the 25 minimal separations are 5 counted five times. Only the 31 product countermodels are 31 distinct facts. That is the shape of the original loop, reproduced here rather than corrected, and it means the published counts carry the information of the smaller ones; the "donor_axis" block of the result artifact carries the table.

The revalidation block cannot see conservativity. Every state it asserts about is built with the donor verdict pinned valid, so the derived rule with its donor conjunct deleted is accepted unchanged. Its acceptance of the derivation is evidence for the non-compensatory primitive only.

5 An axiom set that was not minimal

The interpretation was first stated with four frame conditions. A check that drops each condition and re-runs the proofs reported two of them inert: dropping either lost no theorem — where "lost" then meant "no longer proved", which is the wrong criterion and is corrected below. Neither was a reprieve. Dropping *both* loses three theorems, because each was derivable from the other three, so the four were a redundant presentation of the same constraint. The interpretation now rests on three independent conditions — every coordinate supports the certificate, every edge runs from a coordinate to the certificate, and the certificate is not one of its own coordinates — and the fourth, that the certificate supports nothing, is discharged as a theorem about them. Each of the three loses at least one theorem when dropped, and which theorem each carries is recorded.

An axiom that no theorem needs is either decoration or a redundant presentation of an axiom that is doing the work, and the two are indistinguishable from inside. The check that told them apart is worth more here than the axiom it removed.

The criterion was corrected in stages because each earlier version could turn a fact about search into a claim about an axiom. Counting any theorem that stopped being *proved* credited an axiom when the solver returned `unknown`. Requiring an actual countermodel removed that error, but an open model search was unstable. Bounding discovery to at most four nodes did not remove the instability: CI run 32927946106 reported the edge restriction only intermittently, and run 32946736266 returned `unknown` for its known one-node witness under full-suite load, cascading to four P6 test failures. Both adverse runs remain failure provenance; neither showed the condition inert.

The current measurement no longer discovers a witness. It verifies one pinned, fully specified exact-domain countermodel for each condition. Without coordinate support, a two-node world has an unchanged non-coordinate certificate, one changed coordinate, identity reachability and no edges; this refutes withdrawal by any damage. Without the edge restriction, a one-node world has a non-coordinate certificate with a self-edge; this refutes the derived claim that the certificate supports nothing. Without certificate non-coordinateness, a one-node world makes the changed certificate itself a coordinate with a self-edge; because changed nodes are not reopened, this again refutes withdrawal. All three tables use reflexive reachability and zero ranks. The solver checks the remaining axioms, the exact domain, every table entry and the negated named theorem. A certificate is credited only when every repeated check returns an actual countermodel; `unknown` remains `CANNOT_CHECK`, and an invalid supplied model is a failed certificate rather than evidence of inertness.

These certificates establish local logical necessity relative to this formalization and theorem set only. They were written and checked in the same lane. External or independent validation remains `CANNOT_CHECK` under P6-U-T4; the certificates provide neither empirical evidence nor external scientific authority.

6 Bounded general statement

> Strong execution, action, workflow and attestation certificates can be composed and preserved as lower-level objects across change in a dynamic scientific state, but preservation of *scientific* standing requires an explicit lift across claim-specific semantic obligations; a scientifically material change reopens exactly the coordinates reachable from it, and revalidating precisely those restores the lift while any proper subset of them does not.

The separation and propagation halves of that statement hold at any size. The five registered lift coordinates and five donor families are an instance of it, recovered rather than assumed.

7 Limits

The lift coordinates are not claimed to be universally minimal. No claim is made that existing certificate systems fail their native goals, or that any formal result here establishes utility for a deployed system; nothing in this paper is an empirical claim.

Two limits travel with the mechanization. Reachability is axiomatised rather than defined, because transitive closure is not first-order definable, so the theorems are relative to that axiomatisation — the well-founded rank included. And the interpretation connecting the bounded model to the general theorems was written alongside the model it interprets; its frame conditions are shown to be load-bearing and its counts are produced by an implementation checked against the theorems, but no reviewer outside that lane has examined either — and, as recorded above, those counts are consistent with three graphs other than the one claimed. Independent proof review remains outstanding, and the withdrawal above of a "second implementation" that could not disagree is the reason to be exact about it.

Naming what stands in review's place is the other half of that disclosure. The mechanized results are only as strong as the code that had to be correct for them to hold, and that code is not hypothetical: it is five in-repository kernel modules — `orion.core.claims`, `orion.programme.guard_exercise`, `orion.programme.identity`, `orion.programme.records` and `orion.programme.refutation_capacity` — together with the Z3 solver the checks defer to. Z3 is not reviewed here and its correctness is assumed. This list is not written by hand; it is the transitive import closure of the formal executables, recomputed by `orion.programme.p6_trusted_base`, which fails closed if the disclosure and the imports diverge. A trusted base stated once and left to drift would be a worse disclosure than none, because it would read as current.

Finally, the naturalistic question is untouched. No multi-domain corpus of real changes — code patches, dataset revisions, model revisions, analysis changes, instrument recalibrations, manuscript claim revisions — has been assembled, so no claim is made about revalidation cost saved against revalidating everything or against dependency-graph-only propagation.

8 Conclusion

The strongest version of this result is not a rejection of existing certificate systems; it is a way to use them more completely, and V5's contribution is that the way is now proved rather than enumerated. Runtime proof, certified traces, portable action receipts, workflow signatures, provenance and execution attestation remain valuable and reusable after absorption into a scientific stack. What the lifting semantics adds is a typed bridge from operational certificates to scientific standing, together with soundness, completeness and minimality results for what a change reopens — at any size, with the bounded model of V4 as one instance.

Three of V4's seven reported quantities are withdrawn here, and the check that was supposed to validate change propagation is shown to have asserted tautologies. The result is a smaller set of claims held more firmly. That exchange is the point of the exercise.

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